Absorption and Elimination of Oat Avenanthramides (AVA) in Humans after Acute Consumption of Oat Cookies

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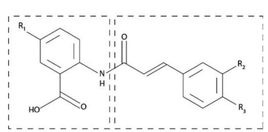
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Abstract

Background: Avenanthramides (AVA) are a group of diphenolic acids found only in oats that have anti-inflammatory and antioxidant effects. Although previous studies report the bioavailability of AVAs in humans, whether AVAs are bioavailable in humans after acute consumption of oat cookies made with natural oat flour is unknown. **Objective:** To examine the metabolic fate of orally ingested oat AVAs by measuring plasma AVA concentrations and pharmacokinetic properties. **Methods:** Male and female non-obese participants (n=16; (ClinicalTrials.gov identifier: NCT02415374) consumed three cookies made with oat flour containing high (229.6 mg/kg, H-AVA) or low (32.7 mg/kg, L-AVA) amounts of AVAs, including AVA-A, AVA-B, and AVA-C compounds. Participants consumed the cookies in the morning after a 10-h fast, and blood samples were collected at 0, 0.5, 1, 2, 3, 5, and 10 h. Plasma total (conjugated and free) AVA concentrations were quantified using ultra-performance liquid chromatography quadrupole time-of-flight mass spectrometry, and pharmacokinetic parameters were calculated. **Results:** After participants consumed the oat cookies, AVA-A, AVA-B, and AVA-C were present at peak concentrations in plasma between 2 and 3 h for the H-AVA group and between 1 and 2 h for the L-AVA group. Maximal plasma concentrations for AVA-A, AVA-B, and AVA-C were higher in the H-AVA than in the L-AVA group. AVA-B had a longer half-life and slower elimination rate than AVA-A and AVA-C. **Conclusions:** AVAs found naturally in oats are bioavailable in humans. AVA-B has the slowest elimination rate and highest half-life compared to AVA-A and AVA-C, while AVA-C has the lowest absorption of all three AVAs.. Further studies are needed to determine the mechanisms contributing to the different absorption profiles of AVAs.

Introduction

Avenanthramides (AVAs) are a group of diphenolic acids found only in oats (*Avena sativa*) (1). Of all the AVAs identified, AVA-A (2p), AVA-B (2f), and AVA-C (2c) are most abundant and differ only by a single moiety on the hydroxycinnamic acid ring (Figure 1). Early research showed that AVAs have the anti-inflammatory and antioxidant effects to prevent cardiovascular diseases and muscle injuries. AVA bioavailability has been demonstrated in animals and humans. Serum levels of AVA 2p, 2f, and 2c reach a peak 2 h after humans consume an AVA-enriched mixture (AEM) and then gradually return to resting levels after 10 h (2). Rats orally gavaged with a mixture of synthetic AVAs show significant quantities of all three AVA fractions in the plasma, heart, liver, and skeletal muscle during a 48-h period (3). Although previous studies report that AVAs are bioavailable and detectable in humans, these studies utilized an AEM or specially treated oats, which might contain much higher amounts of AVAs than regular oats. Here, we examined the metabolic fate of orally ingested oat AVAs by measuring plasma concentrations and pharmacokinetic properties of AVAs. Our results provide the first evidence of AVA bioavailability in humans after acute consumption of oat cookies made with natural oat flour.



Compound	R_1	R_2	R_3	Anthranilic acid	Cinnamic acid
AVA-A	OH	H	OH	5-Hydroxyanthranilic acid	*p*-Coumaric acid
AVA-B	OH	OCH_3	OH	5-Hydroxyanthranilic acid	Ferulic acid
AVA-C	OH	OH	OH	5-Hydroxyanthranilic acid	Caffeic acid
Tranilast	H	OCH_3	OCH_3	Anthranilic acid	Dimethyl caffeic acid

Figure 1. Chemical structure of Avenanthramides

Table 1. AVA content in H-AVA and L-AVA cookies

	AVAs	One cookie (55 g)	Per 100 g	Per serving (3 cookies, 165 g)
H-AVA	Total	7.20	13.10	21.61
	A (mg)	1.70	3.09	5.09
	B (mg)	2.89	5.25	8.67
	C (mg)	2.62	4.76	7.85
L-AVA	Total	1.03	1.86	3.08
	A (mg)	0.19	0.35	0.57
	B (mg)	0.32	0.58	0.95
	C (mg)	0.52	0.94	1.55

Figure 2. Testing oat cookies



Table 2. Pharmacokinetic properties of AVAs

	C_{max} (ng/ml)		T_{max} (h)		$T_{1/2}$ (h)	
	H-AVA	L-AVA	H-AVA	L-AVA	H-AVA	L-AVA
AVA-A	8.39±4.2^{c}	1.98±1.3*	2.50±1.2	1.80±1.3	2.16±0.6^{b}	2.44±0.8
AVA-B	8.44±2.3^{c}	2.43±1.1*	2.04±0.9	1.50±1.5	4.23±0.8^{ac}	4.60±2.7
AVA-C	4.26±2.0^{ab}	1.33±0.7*	2.29±1.0	1.32±1.0	2.71±1.2^{b}	2.87±1.4

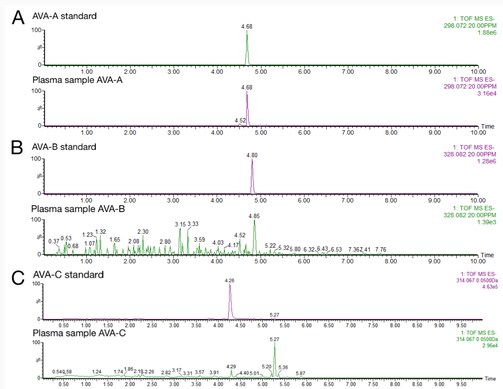
Notes: *P<0.001 vs. H-AVA; ^{a}P<0.05 vs. AVA-A within H-AVA or L-AVA group; ^{b}P<0.05 vs. AVA-B within H-AVA or L-AVA group; ^{c}P<0.05 vs. AVA-C within H-AVA or L-AVA group.

Table 3. Plasma absorption of AVAs

	Mean AUC_{0-t} (h×ng/ml)	Mean K_{el} (h^{-1})	Dose (ng×10^6)
	H-AVA		
AVA-A	42.56±17.9^{bc}	0.35±0.1^{b}	5.09
AVA-B	53.17±12.1^{ac}	0.17±0.1^{ac}	8.67
AVA-C	19.81±12.7^{ab}	0.30±0.1^{b}	7.85
	L-AVA		
AVA-A	8.50±4.4*^{b}	0.32±0.1^{b}	0.57
AVA-B	6.31±1.6*^{ac}	0.20±0.1^{ac}	0.95
AVA-C	3.88±2.0*^{b}	0.30±0.2^{b}	1.55

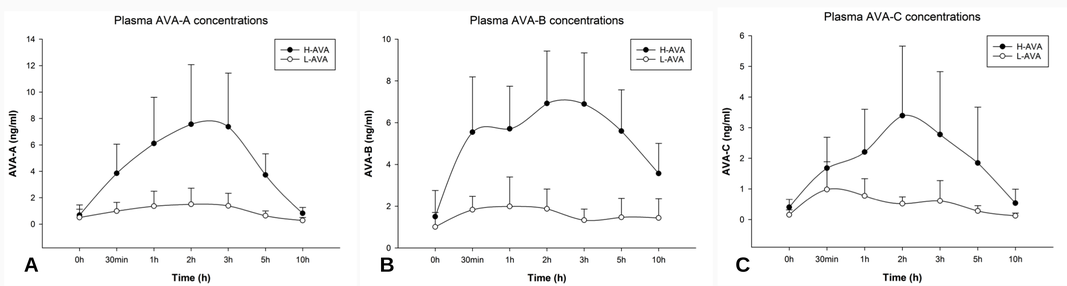
Notes: *P<0.001 vs. H-AVA; ^{a}P<0.05 vs. AVA-A within H-AVA or L-AVA group; ^{b}P<0.05 vs. AVA-B within H-AVA or L-AVA group; ^{c}P<0.05 vs. AVA-C within H-AVA or L-AVA group.

Figure 3. AVAs TOF-MS chromatographs



Notes: Plasma AVA (bottom) and AVA standard (top) TOF-MS chromatographs. A, B and C represent AVA-A, AVA-B and AVA-C respectively.

Figure 4. Changes in plasma AVA concentration over time.



Notes: A, B and C represent AVA-A, AVA-B and AVA-C respectively.

Methods

8 Male and 8 female non-obese participants were recruited in the Twin Cities community (age, 25.6±3.2 years; BMI, 22.6±2.8 kg/m^2). All participants avoided oat consumption 2 days prior to and through the testing period and to consume a low-flavonoid diet 1 week prior to the test. Participants received three cookies (Figure 2) made with oat flour containing a high (H-AVA) or a low amount of AVA (L-AVA) (Table 1). Blood samples were drawn at 0 (before consumption), 0.5, 1, 2, 3, 5 and 10h. AVA was extracted from plasma and concentrations of total AVA in plasma were measured by UPLC-QTOF-MS. Plasma maximal concentration (C_{max}) and time to reach maximal concentration (T_{max}) of AVAs were calculated. Half-life (i.e., time taken for concentration to decrease to half the initial value; $T_{1/2}$) was calculated as a logarithmic concentration. Area under the curve (AUC) for each participant was calculated using the trapezoid method, and mean AUC_{0-t} was presented as mean±SD. Elimination constant (K_{el}) was calculated using K_{el}=0.693/$T_{1/2}$. The mean±SD of plasma total (conjugated and free) AVA was calculated for each time point and plotted against time. AUC_{0-t}, C_{max}, T_{max}, and $T_{1/2}$ for each AVA were compared between H-AVA and L-AVA groups using *t*-tests and among AVA-A, AVA-B, and AVA-C compounds using one-way analysis of variance (ANOVA). P<0.05 was considered statistically significant.

Results

Representative retention times for AVA-A, AVA-B, and AVA-C in standards as measured by Q-TOF-MS were 4.68 min (Figure 3A), 4.80 min (Figure 3B), and 4.26 min (Figure 3C). AVA-A and AVA-C concentrations reached maximum values at 2.50±1.2 h and 2.29±1.0 h in the H-AVA group, and at 1.80±1.3 h and 1.32±1.0 h in the L-AVA group, both returning to baseline levels after 10 h (Figure 4A, 4C). AVA-B concentrations reached maximum values at 2.04±0.9 h in the H-AVA group and 1.50±1.5 h in the L-AVA group but did not return to baseline levels at 10 h (Figure 4B). AVA-B had longer $T_{1/2}$ than AVA-A and AVA-C in the H-AVA group. Also, AVA-A and AVA-B had higher C_{max} than AVA-C in the H-AVA group, and participants in the H-AVA group had higher C_{max} of AVA-A, AVA-B, and AVA-C than participants in the L-AVA group (Table 2). Participants in the H-AVA group had higher doses of and larger mean AUC_{0-t} for AVA-A, AVA-B, and AVA-C compared with participants in the L-AVA group. AVA-C had lowest AUC_{0-t} among three AVAs no matter in H-AVA or L-AVA groups. Similar to the $T_{1/2}$ results, AVA-B had a slower K_{el} than AVA-A or AVA-C in both H-AVA and L-AVA groups (Table 3).

Conclusions

We found that AVAs from cookies made with natural oat flour are bioavailable in humans. AVA-B had the slowest elimination rate and highest $T_{1/2}$. Also, AVA-C had the lowest absorption compared to AVA-A and AVA-B. Further investigations are needed to determine the bioavailability of other AVA compounds and the mechanisms contributing to the different absorption profiles of various AVAs.

References

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